Provenance of the Heat-Flow Material Library

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The library

`matlibs/heatlib_fixed.dat` is a heat-flow material library for FEMM, 136 entries. Each entry carries a thermal conductivity k (W/m K), as a constant or as a table of k against temperature, and a volumetric heat capacity K_t (MJ/m^3 K).

Heat capacity. Every K_t is documented row by row on the FEMM wiki's "Specific Heat" page [1], which names its source; the table's K_t column shows it. All agree with the page to 0.5%.

Thermal conductivity: source. The page does not cover k. It comes from F. M. White's *Heat and Mass Transfer* [8], Appendices C (metals), D (nonmetals), E (liquid metals) and F (saturated liquids): the library's points at 273, 473, 673… K are White's 0, 200, 400… °C columns, and 115 entries take their k from White, point for point, except those described below that take it from another source. Two entries hold fewer points than White (Fireclay brick, engine oil), within 4% of White when interpolated. The "k src." column gives White's appendix. The gases were not compared (White's gas appendix was not among the pages scanned), 6061-T6 and 7075-T6 (not in White) match Lienhard's Table A.1 exactly, and the other aluminium tempers (2024-T3, -O, -T351, -T4, 6061-T651, 7075-T651), HyMu 80, Carpenter 49 and 17-4 PH have no k source recorded.

Thermal conductivity: checks. Independently of White, k was checked against Lienhard's *A Heat Transfer Textbook*, 6th ed. [2], against the *CRC Handbook of Chemistry and Physics* [3], and against two older handbooks [4, 5]. Of the 136 entries, 65 agree with a check source (the worst difference over their points is given), 9 take k from Lienhard or CRC instead of White ("replaced"), 16 are mixed (one check source agrees within 10%, another does not), 18 disagree with every check source by more than 10%, and 28 have no independent check.

Where White's k is wrong or outdated, another source is used. Three cases are errors in White's own table: White gives constantan 26 at 200 °C, which is Lienhard's 100 °C value (Lienhard: 35); it drops 1% carbon steel's 600 °C value, so that its 800 and 1000 °C values sit a column early; and it gives solid silver 355 at 1000 °C, above silver's melting point (silver keeps White's other points and omits that one). For these Lienhard's values are used. White's R-12 and liquid-ammonia data are older than Lienhard's, and Lienhard's are used; White's mercury k (7.3–14.2) is also older, and Lienhard's is used. Three more take their values from the CRC Handbook [3]: Diatomaceous Earth (White's 1.3 is more than ten times the 0.05–0.16 that CRC gives for diatomite and Incropera for the powders), Teflon (0.35; CRC gives 0.26 for PTFE) and Window glass (0.84; CRC gives 1.00 at 0 °C for a window-type soda-lime glass).

The CRC Handbook is the best sourced of these. Its metals table is Ho, Powell and Liley's recommended values (J. Phys. Chem. Ref. Data, 1972) for high-purity specimens; its gas table is from modern reference correlations (Huber and Harvey), and every library gas agrees with it within 6%; its liquids table agrees within 2% for water, ethanol, ethylene glycol and glycerin. Its ceramics table calls itself "only … a rough guide". Where a library entry has only 0 and 200 °C points (silicon, germanium, the diamonds, magnesium oxide), FEMM interpolates linearly across a conductivity that falls roughly as $1/T$, so it overestimates at 400 K by 12–22% against CRC even where the tabulated points agree.

The two older handbooks are secondary compilations of older data and only flag entries; a disagreement means White and they differ, not that White was copied wrongly. Rohsenow's element and alloy tables (2.32, 2.34) are round values in kcal/(m h °C) converted to W/m K, and several of their pure-metal values sit well below modern ones (rhodium 88 against 150, sodium 100 against 134). Its Table 2.35 is Incropera and DeWitt's Table A.3 reprinted, with some decimal slips in the reprint (ice 0.188, paper 0.011 W/m K), which were not used. Many disagreements are a choice of variety rather than an error ("Concrete", "Window" glass, "Sandstone"). Besides the three replaced from CRC, the ones where White itself may be off are Cobalt (White's 104 against 69–70 in the older handbooks; not in CRC) and Vanadium (White's 31 against Kreith's 60; not in CRC), both left as White gives them; White's footnote warns that nonmetal properties are "subject to large uncertainty (up to 50%)".

Fluids

The heat-flow solver models conduction only. A fluid entry is therefore right only where the fluid cannot move, as in a thin gap or a sealed cavity too small to convect. A fluid that flows or convects carries heat far faster than its conductivity alone allows; it belongs in a convection boundary condition (a heat-transfer coefficient h) on the surface it cools, not in a region of its own. The three fluid folders are named "Liquid Metals (stagnant only)", "Saturated Liquids (stagnant only)" and "Gases at 1 atm (stagnant only)". A library entry carries no comment, so the folder name is the only warning a user sees.

Entries

The k column is the constant value (FEMM uses the table when one is given; "Pts." is its length).

Entry	k	K_t	**Pts.**	K_t **src.**	k **src.**	k **check**	**Detail**
Metallic Solids / Aluminum							
Aluminum, Pure	236	2.45	4	[6]	[8] C	checked	0.0% (Table A.1); CRC 12-219 -1%
Aluminum, 2024-T3	121	2.433	–	[9]	–	checked	CRC 12-221 -7%
Aluminum, 2024-O	193	2.433	–	[9]	–	none	
Aluminum, 2024-T351	121	2.433	–	[9]	–	checked	CRC 12-221 -7%
Aluminum, 2024-T4	121	2.433	–	[9]	–	checked	CRC 12-221 -7%
Aluminum, 2024-T6	171	2.433	2	[9]	[8] C	none	
Aluminum, 6061-T6	167	2.419	–	[9]	–	checked	0.0% (Table A.1)
Aluminum, 6061-T651	167	2.419	–	[9]	–	none	
Aluminum, 7075-T6	130	2.698	–	[9]	–	checked	0.0% (Table A.1)
Aluminum, 7075-T651	130	2.698	–	[9]	–	none	
Metallic Solids							
Beryllium	218	3.402	6	[7]	[8] C	mixed	Rohsenow T2.32 +36% at 273 K; Kreith C.4 -4%
Boron	31.7	2.825	6	[7]	[8] C	none	
Cadmium	104	1.99	2	[7]	[8] C	*disagrees*	Rohsenow T2.32 +12% at 273 K; Kreith C.4 +11% at 298 K
Chromium	95	3.257	6	[6]	[8] C	checked	1.2% (Table A.1); CRC 12-219 -2%
Cobalt	104	3.748	2	[7]	[8] C	*disagrees*	Rohsenow T2.32 +45% at 293 K; Kreith C.4 +46% at 298 K
Metallic Solids / Copper							
Constantan	22	3.658	3	[6]	[8, 2] C	**replaced**	White App. C gives 26 at 200 C, Lienhard's 100 C value; Lienhard's 22/26/35 at 273/373/473 K used (Table A.1); CRC 12-221 +0%
Copper, Pure	401	3.438	6	[6]	[8] C	checked	0.0% (Table A.1); CRC 12-219 +1%
Brass	110	3.281	3	[6]	[8] C	checked	3.8% (Table A.1); CRC 12-221 +0%
Bronze	49	3.696	3	[8]	[8] C	none	
Metallic Solids / Core-materials							
HyMu80	34.6	4.321	–	[13]	–	none	
Carpenter49	13	4.125	–	[13]	–	none	
Metallic Solids							
Germanium	67	1.715	4	[7]	[8] C	*disagrees*	CRC 12-219 +15% at 400 K

Entry	k	K_t	**Pts.**	K_t **src.**	k **src.**	k **check**	**Detail**
Gold	318	2.492	6	[6]	[8] C	checked	3.4% (Table A.1); CRC 12-219 +3%
Metallic Solids / Iron							
Iron, Wrought	59	3.542	4	[7]	[8] C	*disagrees*	Rohsenow T2.32 -11% at 873 K
Iron, Pure	83	3.53	6	[6]	[8] C	checked	6.7% (Table A.1; edition difference); CRC 12-219 +9%
Iron, Cast (4% C)	52	3.054	5	[6]	[8] C	checked	0.0% (Table A.1)
Iron, Armco	75	3.634	6	[23, 8]	[8] C	checked	Rohsenow T2.32 +2%; CRC 12-221 +4%
Metallic Solids / Steels							
Steel, 1% Chrome	62	3.616	5	[7]	[8] C	none	
Steel, 1% Carbon	43	3.69	6	[6]	[8, 2] C	**replaced**	White App. C drops the 600 C value, putting 29 and 28 a column early; Lienhard's 43/42/36/33/29/28 at 273-1273 K used (Table A.1); CRC 12-221 -4%
347 Stainless Steel	14	3.829	6	[8]	[8] C	none	
304 Stainless Steel	14	3.768	5	[8]	[8] C	checked	Rohsenow T2.34 +7%; CRC 12-221 +3%
SST 17Cr-4Ni, Cond H900	17.9	3.268	4	[9]	–	none	
Metallic Solids							
Lead	36	1.478	2	[6]	[8] C	checked	0.0% (Table A.1); CRC 12-219 +1%
Lithium	79	1.863	–	[7]	[8] C	*disagrees*	Rohsenow T2.32 +11% at 273 K
Magnesium	157	1.786	4	[6]	[8] C	checked	0.7% (Table A.1); CRC 12-219 -0%
Manganese	7.7	3.559	–	[7]	[8] C	*disagrees*	Rohsenow T2.32 -85% at 273 K
Molybdenum	139	2.565	6	[6]	[8] C	checked	0.0% (Table A.1)
Metallic Solids / Nickel							
Nickel, Pure	94	3.963	6	[6]	[8] C	checked	1.6% (Table A.1); CRC 12-219 +4%
Nichrome	12	3.696	4	[6]	[8] C	checked	Rohsenow T2.34 -5%; CRC 12-221 -4%
Inconel-X	11	3.761	6	[6]	[8] C	checked	2.7% (Table A.1)
Metallic Solids / Platinum							
Platinum, Pure	72	2.853	6	[6]	[8] C	checked	1.3% (Table A.1); CRC 12-219 +1%
60% Pt, 40% Rh	46	2.694	6	[11]	[8] C	none	
Metallic Solids							
Rhenium	49	2.898	6	[7]	[8] C	none	
Rhodium	151	3.013	6	[7]	[8] C	mixed	Rohsenow T2.32 +82% at 373 K; Kreith C.4 -0%
Silicon	168	1.644	6	[6]	[8] C	checked	13% at 200 C (Table A.1; edition difference); CRC 12-219 +22% at 400 K

Entry	k	K_t	**Pts.**	K_t **src.**	k **src.**	k **check**	**Detail**
Silver	428	2.484	5	[6]	[8] C	checked	White App. C without its 1000 C point (355, a solid above the 962 C melting point); within 4% (Table A.1); CRC 12-219 -2%
Sodium	135	1.177	–	[7]	[8] C	mixed	Rohsenow T2.32 +45% at 323 K; Kreith C.4 +1%
Tin	68	1.659	2	[11]	[8] C	checked	0.0% (Table A.1); CRC 12-219 +1%
Titanium	22	2.374	6	[6]	[8] C	checked	4.8% (Table A.1); CRC 12-219 +2%
Tungsten	182	2.574	6	[6]	[8] C	checked	3.3% (Table A.1); CRC 12-219 +3%
Uranium	27	2.169	5	[6]	[8] C	checked	4.9% (Table A.1)
Vanadium	31	2.806	6	[7]	[8] C	*disagrees*	Kreith C.4 -48% at 298 K
Zinc	122	2.772	3	[6]	[8] C	checked	2.0% (Table A.1)
Zirconium	23	1.853	6	[7]	[8] C	none	
Nonmetallic Solids							
Aluminum Oxide	40	3.038	6	[6]	[8] D	mixed	CRC 12-222 +19% at 373 K; CRC 12-224 +3%
Asbestos	0.11	0.313	–	[8]	[8] D	*disagrees*	CRC 12-224 +22% at 273 K
Beryllium Oxide	302	3.09	6	[8]	[8] D	mixed	CRC 12-222 +10%; CRC 12-224 +50% at 1273 K
Nonmetallic Solids / Bricks							
Brick, Common	0.7	1.344	–	[10]	[8] D	checked	Rohsenow T2.35 -3%; Kreith C.1 -1%
Brick, Chrome	2.2	2.52	5	[10]	[8] D	checked	Rohsenow T2.35 -4%
Brick, Fireclay	1	1.449	2	[7]	[8] D	checked	Rohsenow T2.35 +5%; CRC 12-224 -8%
Brick, Magnesite	4	1.92	6	[10]	[8] D	*disagrees*	Rohsenow T2.35 -17% at 922 K
Brick, Masonry	0.66	1.674	–	[7]	[8] D	none	
Brick, Silica	1.1	1.355	–	[14]	[8] D	none	
Nonmetallic Solids							
Carbon	1.6	1.602	6	[16]	[8] D	none	
Cement Mortar	0.9	1.451	–	[8]	[8] D	*disagrees*	Rohsenow T2.35 +25% at 300 K; CRC 12-224 +64% at 363 K
Concrete	1	1.511	–	[7]	[8] D	*disagrees*	Rohsenow T2.35 -29% at 300 K; Kreith C.1 -41% at 300 K; CRC 12-224 +25% at 273 K
Coal	0.24	1.625	–	[10]	[8] D	checked	CRC 12-224 -8%
Nonmetallic Solids / Diamonds							
Diamond Type I	1000	1.776	2	[7]	[8] D	*disagrees*	CRC 12-219 -15% at 400 K
Diamond Type IIa	2650	1.782	2	[10]	[8] D	*disagrees*	CRC 12-219 +16% at 400 K
Diamond Type IIb	1510	1.658	2	[6]	[8] D	*disagrees*	CRC 12-219 +12% at 400 K
Nonmetallic Solids / Earths							
Clay	1.4	1.285	–	[12]	[8] D	checked	Rohsenow T2.35 +8%

Entry	k	K_t	**Pts.**	K_t **src.**	k **src.**	k **check**	**Detail**
Diatomaceous Earth	0.09	0.35	2	[7]	[8, 3] D	**replaced**	from CRC 12-224 diatomite, 0.5 g/cm3: 0.09 and 0.16 at 273 and 673 K; White's 1.3 is ten times diatomite powders' 0.05-0.16; Rohsenow T2.35 +37% at 300 K; CRC 12-224 +0%
Sand	0.3	1.205	–	[7]	[8] D	checked	CRC 12-224 -9%
Nonmetallic Solids / Glasses							
Corning Pyrex 7740	1.1	1.616	4	[7]	[8] D	mixed	Rohsenow T2.35 -19% at 300 K; CRC 12-226 -4%
Pyroceram	4.1	2.033	6	[7]	[8] D	none	
Window	1	2.268	2	[10]	[8, 3] D	**replaced**	from CRC 12-227 soda-lime glass (72 SiO2, 15 Na2O, 11 CaO, 2 Al2O3): 1.00 and 1.15 at 273 and 373 K; White gives 0.84; Rohsenow T2.35 -26% at 300 K; Kreith C.1 +8%; CRC 12-226 +0%
Nonmetallic Solids							
Ice	2.2	1.756	–	[10]	[8] D	checked	Kreith C.1 +0%; CRC 12-222 +0%
Nonmetallic Solids / Insulations							
Cork, granular	0.045	0.301	–	[7]	[8] D	checked	Rohsenow T2.35 +0%
Corkboard	0.043	0.304	–	[8]	[8] D	mixed	Rohsenow T2.35 +10% at 300 K; Kreith C.1 +7%
Cellulose, loose	0.038	0.093	–	[15]	[8] D	checked	Rohsenow T2.35 -3%
Feltboard	0.035	0.179	–	[17]	[8] D	none	
Glass Fiber	0.035	0.02688	–	[10]	[8] D	checked	Rohsenow T2.35 -3%
Glass Wool	0.038	0.0168	–	[10]	[8] D	mixed	Rohsenow T2.35 +0%; CRC 12-224 -24% at 373 K
Kapok	0.035	0.207	–	[18]	[8] D	none	
Magnesia, 85%	0.065	0.179	2	[19]	[8] D	mixed	Rohsenow T2.35 +35% at 365 K; Kreith C.1 -5%; CRC 12-224 -15% at 303 K
Polystyrene	0.025	0.05198	–	[7]	[8] D	checked	Rohsenow T2.35 -7%
Rubber, Foam	0.03	0.2092	–	[7]	[8] D	mixed	Rohsenow T2.35 -6%; CRC 12-224 -40% at 293 K
Rock Wool	0.04	0.084	–	[20]	[8] D	checked	Kreith C.1 +0%; CRC 12-224 +0%
Sawdust	0.059	0.336	–	[15]	[8] D	mixed	Kreith C.1 -26% at 300 K; CRC 12-224 -2%
Vermiculite, loose	0.058	0.1657	–	[7]	[8] D	mixed	Rohsenow T2.35 -15% at 300 K; Kreith C.1 +0%
Nonmetallic Solids							
Magnesium Oxide	53	3.15	6	[6]	[8] D	*disagrees*	CRC 12-224 +14% at 373 K
Plaster	0.5	1.36	–	[7]	[8] D	none	
Quartz, fused	1.3	1.639	6	[7]	[8] D	checked	CRC 12-222 -9%; CRC 12-226 -3%
Nonmetallic Solids / Rocks							

Entry	k	K_t	**Pts.**	K_t **src.**	k **src.**	k **check**	**Detail**
Granite	3	2.165	–	[6]	[8] D	*disagrees*	CRC 12-224 +36% at 293 K
Limestone	2	2.07	–	[6]	[8] D	mixed	Kreith C.1 +5%; CRC 12-224 +100% at 293 K
Marble	2.7	2.65	–	[8]	[8] D	checked	Kreith C.1 +4%
Sandstone	2.8	1.628	–	[8]	[8] D	*disagrees*	Kreith C.1 +65% at 300 K; CRC 12-224 +115% at 293 K
Nonmetallic Solids							
Rubber, hard	0.16	2.4	–	[10]	[8] D	checked	CRC 12-224 +0%
Skin, human	0.37	2.666	–	[21]	[8] D	none	
Nonmetallic Solids / Snow							
Snow, loose	0.05	0.2092	–	[7]	[8] D	none	
Snow, packed	0.19	1.151	–	[7]	[8] D	none	
Nonmetallic Solids							
Teflon-min-heat-capacity	0.26	2.58	–	[22]	[8, 3] D	**replaced**	from CRC 12-225 PTFE at 25 C, 0.26; White gives 0.35; CRC 12-224 +0%
Teflon-max-heat-capacity	0.26	3.22	–	[22]	[8, 3] D	**replaced**	from CRC 12-225 PTFE at 25 C, 0.26; White gives 0.35; CRC 12-224 +0%
Nonmetallic Solids / Woods							
Balsa-min-heat-capacity	0.055	0.2991	–	[7]	[8] D	mixed	Kreith C.1 +10%; CRC 12-224 +38% at 303 K
Balsa-max-heat-capacity	0.055	0.7363	–	[7]	[8] D	mixed	Kreith C.1 +10%; CRC 12-224 +38% at 303 K
Fir	0.11	1.632	–	[6]	[8] D	mixed	Rohsenow T2.35 -8%; CRC 12-224 -21% at 293 K
Cypress	0.097	1.058	–	[7]	[8] D	none	
Maple	0.17	1.634	–	[7]	[8] D	checked	Rohsenow T2.35 +6%
Oak	0.17	1.434	–	[6]	[8] D	checked	Rohsenow T2.35 +6%; Kreith C.1 +0%; CRC 12-224 +6%
Pine, yellow	0.15	1.795	–	[12]	[8] D	none	
Pine, white	0.11	1.22	–	[7]	[8] D	checked	Rohsenow T2.35 -8%; Kreith C.1 -8%; CRC 12-224 +0%
Plywood	0.12	0.66	–	[6]	[8] D	checked	Rohsenow T2.35 +0%; CRC 12-224 +9%
Nonmetallic Solids							
Wool	0.038	0.041	–	[15]	[8] D	checked	CRC 12-224 -5%
Liquid Metals (stagnant only)							
Bismuth, liquid	16.8	1.42	6	[7]	[8] E	checked	Rohsenow T2.29 +0%
Lead, liquid	15.5	1.601	5	[7]	[8] E	checked	7.5% at 1000 K (Table A.3; older data)
Mercury	6.97	1.863	7	[7]	[8, 2] E	**replaced**	Lienhard's k, 6.97-12.0 W/m/K (Table A.3); White's App. E k (7.3-14.2) is older; CRC 6-254 -7%
Lithium, liquid	43	2.147	7	[7]	[8] E	none	
Potassium, liquid	55	0.6724	8	[7]	[8] E	*disagrees*	Rohsenow T2.29 +14% at 422 K
Sodium, liquid	88	1.284	8	[7]	[8] E	checked	Rohsenow T2.29 +3%

Entry	k	K_t	**Pts.**	K_t **src.**	k **src.**	k **check**	**Detail**
Saturated Liquids (stagnant only)							
Ammonia	0.586	2.88	5	[6]	[8, 2] F	**replaced**	Lienhard Table A.3 values; White App. F data are older (up to 13% off)
Ethyl Alcohol	0.186	1.89	7	[8]	[8] F	checked	Kreith B.2 -3%; CRC 6-254 +1%
Ethylene Glycol	0.254	2.66	6	[7]	[8] F	checked	Kreith B.2 -0%; CRC 6-254 +2%
Freon-12	0.0909	1.296	5	[6]	[8, 2] F	**replaced**	Lienhard Table A.3 values; White App. F data are older (up to 24% off; wrong trend)
Glycerin	0.282	3.016	4	[8]	[8] F	checked	Kreith B.2 +0%; CRC 6-254 +1%
Unused Engine Oil	0.147	1.669	2	[8]	[8] F	none	
Water	0.564	4.184	12	[7]	[8] F	checked	2.9% (Table A.3); CRC 6-254 +1%
Gases at 1 atm (stagnant only)							
Air	0.0181	0.001185	18	[6]	–	checked	4.1% (Table A.6); points above 1500 K not checkable; CRC 6-252 -2%
Ammonia	0.0153	0.001683	9	[6]	–	checked	6.9% at 700 K (Table A.6); points outside 240-700 K not checkable; CRC 6-252 -5%
Argon	0.0124	8.47×10^{-4}	14	[6]	–	checked	1.5% (Table A.6); points above 700 K not checkable; CRC 6-252 -1%
Carbon Dioxide	0.0095	0.001533	7	[6]	–	checked	1.8% (Table A.6); points outside 220-600 K not checkable; CRC 6-252 -4%
Carbon Monoxide	0.0175	0.001184	6	[6]	–	checked	4.5% (Table A.6); CRC 6-252 -3%
Helium	0.115	8.44×10^{-4}	6	[6]	–	checked	5.3% (Table A.6); CRC 6-252 -5%
Hydrogen	0.128	0.001172	6	[6]	–	checked	7.0% at 1000 K (Table A.6; older data); CRC 6-252 -6%
Nitrogen	0.0183	0.001185	7	[6]	–	checked	4.7% (Table A.6); CRC 6-252 -2%
Oxygen	0.0182	0.001197	7	[6]	–	checked	0.6% (Table A.6); CRC 6-252 -1%
Steam at 373.15 K	0.0264	0.001243	7	[6]	–	checked	10% at 1000 K (Table A.6; older data); CRC 6-252 +1%

References

[1] FEMM wiki, "Specific Heat," `https://www.femm.info/doku/doku.php?id=specificheat`, saved 1 October 2026: the source of every Kt entry, row by row.

[2] J. H. Lienhard IV and J. H. Lienhard V, *A Heat Transfer Textbook*, 6th ed., v6.00, 2024, `https://ahtt.mit.edu`, Tables A.1, A.3 and A.6. Used to check k, and the source of k for the entries marked "replaced" that cite

it.

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